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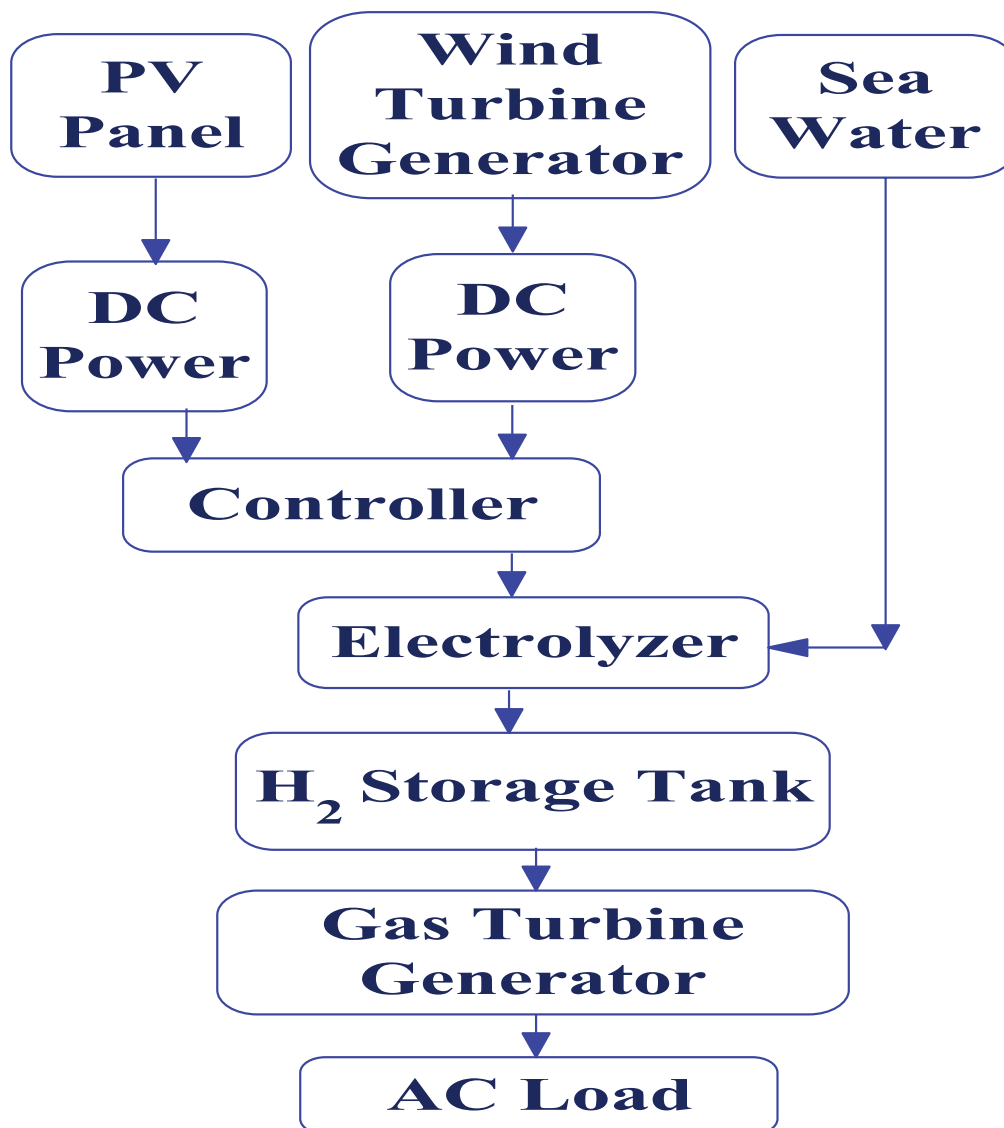
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ELECTRICAL & ELECTRONIC ENGINEERING | RESEARCH ARTICLE

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ELECTRICAL & ELECTRONIC ENGINEERING | RESEARCH ARTICLE

Design & simulation of hydrogen based hybrid green power system using sea water for Cox's Bazar

Kazi Meharajul Kabir^{1*} and Mahmud Abdul Matin Bhuiyan²

Abstract: Hydrogen energy is one of the favorite choice to alternative power generation around the world. Renewable energy is the best scope to produce alternative power. Because resources are available and doesn't make any green house effect. In this reason to establish a renewable resources (solar, wind) based hybrid system with **sea water electrolysis by Proton exchange membrane (PEM) electrolyzer** for the production of Hydrogen. In this proposed system PV & Wind power use for sea water electrolysis and storage Hydrogen. To meet the peak load demand Gas Turbine Generator used to produce electricity using storage Hydrogen fuel. For design, optimization & peak load demand we have chosen Diabatic beach, Kolatoli Cox's Bazar where wind flow, solar radiation and sea water are available.

Subjects: Engineering Management; Power & Energy; Clean Tech; Design; Electrical & Electronic Engineering

Keywords: hybrid system; photovoltaic; wind; sea water; hydrogen; gas turbine generator; innovative design and manufacturing

1. Introduction

A developing country like Bangladesh is experiencing scarcity of efficient electricity production system. Currently, about 72% of the total population has access to electricity including public & private sectors (Ministry of Power, 2016). Main sources of electricity production in Bangladesh are natural gas: it's about 66% and others are oil, hydropower and coal (*An overview of power sector of Bangladesh*, 2011). The gas reserve will be expended in next 10–20 year. In 2016 our total daily generation is 8,900 MW and production capacity 12,725 MW (Bangladesh Power Development Board, 2016). After

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Kazi Meharajul Kabir

PUBLIC INTEREST STATEMENT

At 21st century, the importance of energy demand does not require any explanation. World's major power generation sources fossil fuel are in saturation period. Even though, nuclear power plant is being capable of yielding huge amount of power. However, due to dangers associated with nuclear power plant, electricity production is quite risky for society. So, renewable energy is the best choice for alternating power generation to overcome power crisis and it is environment friendly. In This article proposed a hydrogen sea shore based hybrid power system for Cox's Bazar, Bangladesh for overcame the power crisis. The main objectives of this article is design and simulation of a hybrid system also creating different scope for producing hydrogen fuel because hydrogen fuel is the future fuel for power generation and transport section.

15 year in 2030 our demand will be 30,000 MW. So it's very necessary for us to improve our power generation capacity. To overcome this situation, government has taken necessary steps. For instance, establishing various power plants which are non-renewable; such as oil, gas, and coal based plants, and many more. In addition, government has taken steps to establish a 1,000 MW nuclear power plant in Ruppur. But nuclear power needs a huge technological safety. If any kinds of disaster happen we can't handle it because of our lack of technical ability. Government has taken steps to establish 1,320 MW thermal coal based plant in Rumpal & 1,200 MW plant in Moheshkhali also 1,200 MW plant in Baskhali. It's also harmful for our nature. To meet the growing electricity demand Government also has taken furnace oil based quick rental power plant for quickly solving power crisis but per units cost of this types of power plant is high. It has negative environmental effects. So we can't effort these types of power plant longer period. The power sector alone contributes to 40% of total CO₂ emissions in Bangladesh (Shrestha, Anandarajah, & Liyanage, 2009). So it is very necessary for us to invent new source of energy for solving national power crisis. Therefore, renewable energy is one of the best efficient choices for overcoming this power crisis. As renewable sources are natural and environment friendly and doesn't create any kind of disaster. In Bangladesh, about 72% of the population lives in the rural areas and renewable energy is providing clean energy (Chowdhury, Reza, Nitol, & Mahabub, 2012).

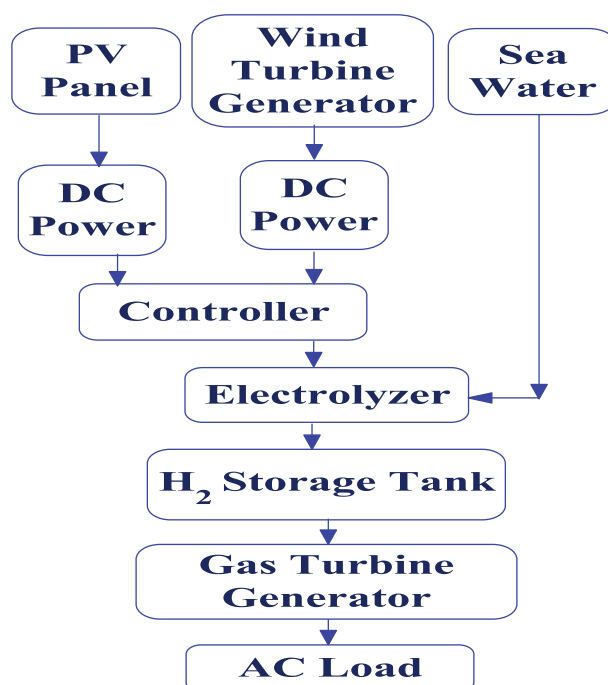
To overcome this situation, hydrogen economy is one of the best solutions to the world's future energy and plays an important role as an energy carrier. In near future Hydrogen may be converted into useful forms of energy more efficiently than fossil fuels (Veziroglu, 1998). In addition, water electrolysis technology is one way to produce hydrogen (*FreedomCAR and fuel partnership*, 2009). Proton exchange membrane (PEM) Electrolyzer becomes known as one of the most clean to produce hydrogen and talented alternatives to reduce fossil fuel dependency (Kunusch, Puleston, Mayosky and Husar, 2011). As hydrogen is a green fuel so there is no (CO₂) carbon emission. In Kolatoli, Cox's Bazar annual average solar radiation of 4.77 kWh/m²/day and rate of average wind speed is 3.55 m/s. Sea water also available in Cox's Bazar because Cox's Bazar one of the coastal areas in Bangladesh To meet the peak load demand in a day which is 9 MW and the demand schedule are 5–10 PM So the total demand in a year is 3,285 MWh/yr (Ambia, Islam, Shoeb, Maruf, & Mohsin, 2010).

2. System methodology

To establish a green energy system we have used **two renewable energy (solar and wind)**. When sun radiation falls on the PV panel, it will produce direct current and when wind flow rotate the turbine connected to DC Generator it also produce dc power too. Both powers will control by a controller for requirement of PEM electrolyzer input power. PEM electrolyzer will produce Hydrogen (H₂) using sea water. Then Hydrogen will Store by a tank. To meet the peak load demand Gas Turbine Generator will use to produce electricity by source of hydrogen (H₂) fuel.

Figure 1 represented the block diagram of the system. The major objective of this proposed system to be able to meet the peak load demand in Kolatoli, Cox's Bazar. HOMER PRO software uses the calculations, analysis, simulations of the system.

Figure 1. Block diagram of the hybrid system.



3. Proposed hybrid green system

The propose hybrid green system consists of two renewable sources (solar, wind), hydrogen storage tank, peak load, gas turbine generator, and sea water which is used for producing hydrogen by electrolyzer.

The proposed hybrid system model & geographical location is situated at 22 degrees 26 min North latitude and 91 degrees 57 min East longitude at Diabatic beach, Cox's Bazar, Bangladesh which is also near the Cox's Bazar international airport. Table 1 Shown proposed system component specification and Figures 2 and 3. Showed proposed hybrid system model and location.

Table 1. Proposed system component specification

Components	Model name	Capacity
PV cell	Generic flat plate PV	2.2 MW
Wind turbine generator	XL-500	4 MW
Gas turbine generator	3 MW Genset	3 MW
Electrolyzer	PEM electrolyzer	3 MW
Storage tank	Hydrogen storage tank	50,000 kg
Project life		25 years

Figure 2. Proposed hybrid system location.

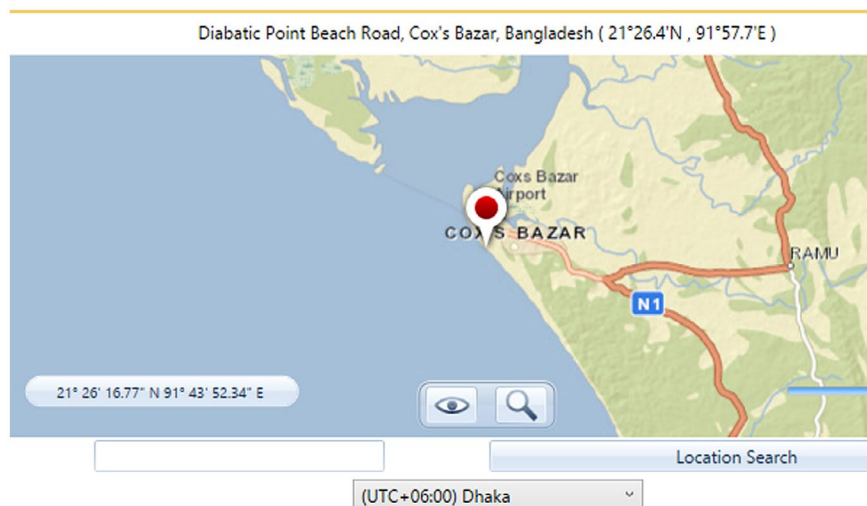
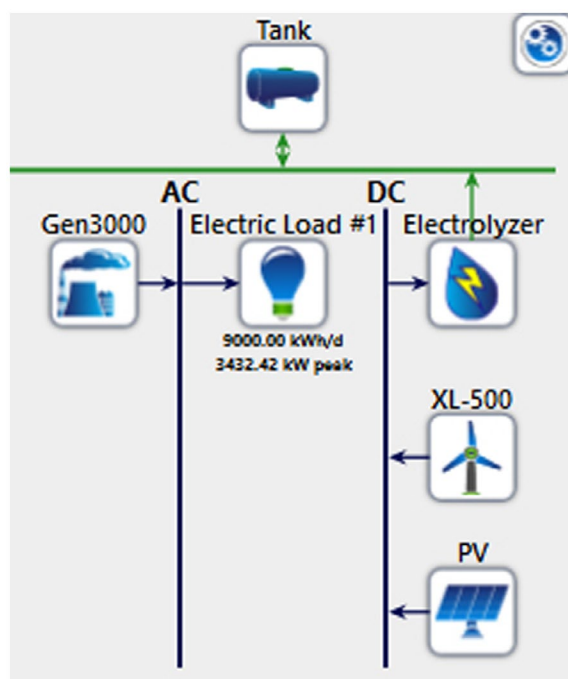


Figure 3. Hybrid system model.



4. Resources analysis

4.1. Photovoltaic energy

PV cell converts light energy into the electrical energy. Sun radiation fall into the panel then PV cell material and electrons are emitted from the atoms then they are outspread. Solar cell released current created solar power converted into electrical power (Power Cell, Ministry of Power, Bangladesh, 2016). PV module converts only 40% of sunlight radiation to Electrical power (Lipu, 2013). The PV system gives DC Power as a output. The PV output power can be calculated according to the following equation:

$$P = Ax^2 + Bx + C \text{ (in Watts)} \quad (1)$$

where, x = solar radiation, P = power generation, and A , B , C are constants, which can be derived from measured data.

4.2. Wind energy

Wind energy converted into electricity by using the kinetic energy into mechanical energy (Lipu, 2013). When air flow comes to wind turbine propellers, its rotated then converts mechanical power into electrical power. According to Betz limit, wind turbine can convert only 59% air energy into electricity. The output for a wind generator is given

$$P = (1/2) \times \rho \times A \times v^3 \text{ (in Watts)} \quad (2)$$

where A = area perpendicular to the direction of flow (in m^2), v = wind velocity (m/s), ρ = density of air (in kg/m^3) and P = power generation.

Maximum power efficiency of any types of wind turbine is 0.59. This is called the “power coefficient”

$$C_{p\max} = 0.59 \quad (3)$$

4.3. Sea water electrolysis

The main purpose of the sea water electrolysis is to produce hydrogen (H_2). PEM Electrolyzer used for producing Hydrogen. Electrolysis of water is its decomposition to give hydrogen and oxygen gases due to the passage of an electric current. The chemical process of electrolysis (Potential of micro hydro power units in Bangladesh, 2012) are given below:

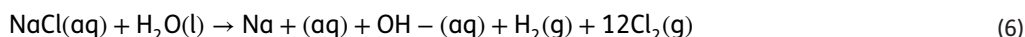
The reaction at the cathode is:



The reaction at the anode is:



The overall reaction is as follows:



Reduction of $\text{Na} + (\text{E}^\circ = -2.7 \text{ V})$ is energetically more difficult than the reduction of water (-1.23 V).

4.4. Data analysis

Cox's Bazar Diabatic beach is located at 21 degrees 26.4 min North latitude and 91 degrees 57.7 min East longitude. One year wind speed and solar radiation collected from NASA (Kabir, Nath, Mazumder, & Islam, 2016; NASA surface meteorology and solar energy, 2016). Table 2 shows the Solar Global Horizontal Irradiance (GHI) value & wind speed for Cox's Bazar.

Table 2. Solar GHI value & wind speed data for Cox's Bazar

Month	Clearness index	NASA Daily radiation (kWh/m ² / day)	NASA Average wind speed (m/s) height (20 m)
January	0.653	4.750	2.90
February	0.641	5.330	3.13
March	0.620	5.930	3.40
April	0.579	6.090	3.68
May	0.503	5.520	3.75
June	0.372	4.110	4.87
July	0.347	3.810	4.83
August	0.380	4.030	4.32
September	0.424	4.150	3.35
October	0.527	4.530	2.70
November	0.601	4.480	2.86
December	0.659	4.560	2.82
Annual average	0.526	4.77	3.55

5. Simulation & results

5.1. System component specifications for simulation

For simulate and calculate the hybrid system using HOMER Pro software (HOMER energy, 2016; Lambert, Gilman, & Lilienthal, 2006).

The proposed solar panel capacity is 2,200 kW which is shown in Table 3 and Figure 4 presents the solar radiation data.

The wind turbine capacity (500 kw × 9) = 9 kW which is shown in Table 4 and Figure 5 presents wind speed data.

The capacity of storage tank is 50,000 kg and PEM electrolyzer capacity is 300 kW which shown in Table 5. The capacity of gas turbine generator is 3,000 kw and uses fuel is hydrogen, fuel lower heat-ing value is 120 MJ/kg which is shown in Table 6. Electrical peak load demand in patenga is 9 MW and demand duration are 6–10 PM which shown in Table 7.

5.2. Simulation results

The rated capacities of the PV Cells are 2.2 MW and total electricity production is 3,263.389 MWh/yr. The mean output is 0.373 MW. The per unit cost of energy of the PV system is 0.039\$/kWh which is shown in Table 8 and hourly electricity production curve showed in Figure 6.

Table 3. PV cell specification

Solar cell	2,200 kW
Lifetime	20 years
Derating factor	80%
Tracking system	No tracking
Slope	23.00°
Ground reflectance	20%

Figure 4. Solar radiation curve.

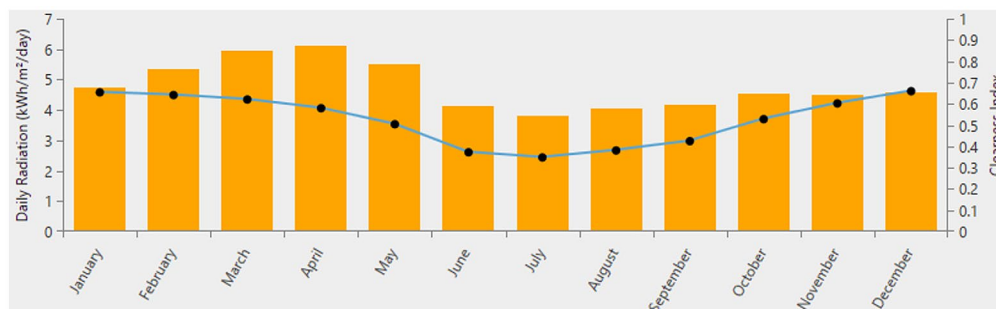


Table 4. Wind turbine generator specification

Wind turbine	9
Rated capacity	500 kW × 9 = 4.5 MW
Rotor diameter	22 m
Hub height	35 m
Start up speed	2.5 m/s
Cut in speed	3 m/s
Cut of speed	25 m/s
Life time	20 years

Figure 5. Wind speed curve.

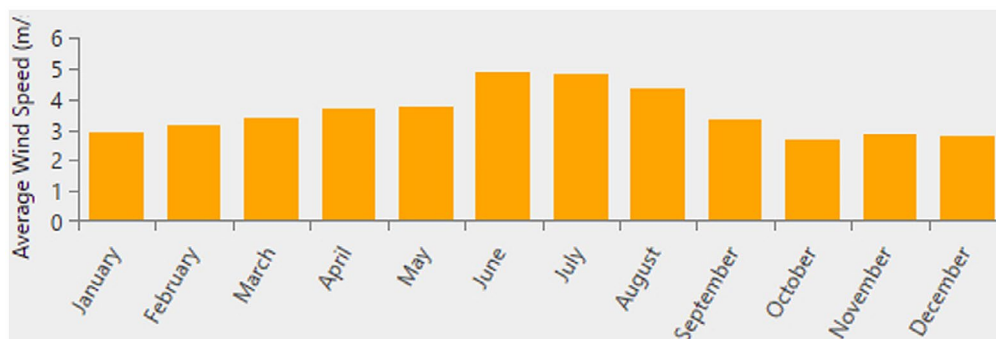


Table 5. Storage tank & electrolyzer specification

Hydrogen storage tank	50,000 Kg
PEM electrolyzer	3,000 KW
Life time	15 yr
Efficiency	85%

Table 6. Gas turbine generator

Capacity	3,000 KW
Lifetime	25,000 h
Fuel used	Hydrogen
Fuel curve intercept	0.01400 L/hr/kW
Fuel curve slope	0.0420 L/hr/kW

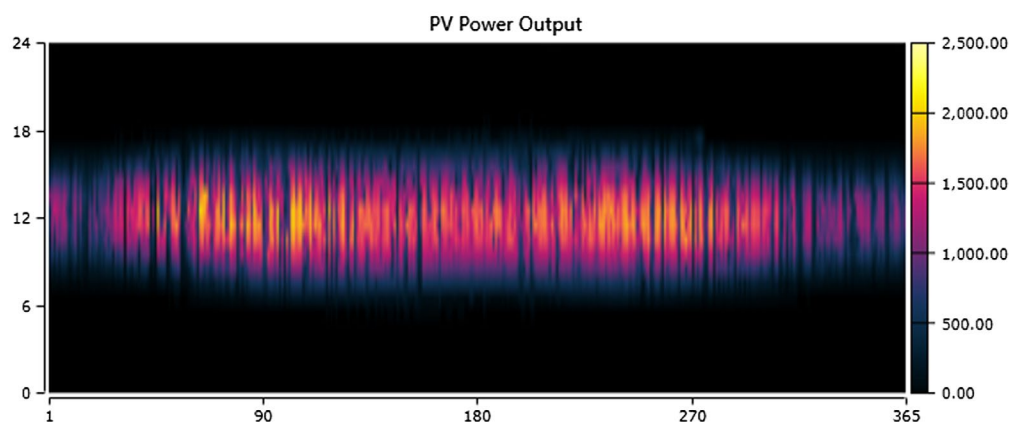
Table 7. Electrical peak load demand

Load type	AC
Demand schedule 5–6 PM	1,500 kWh
Demand schedule 6–7 PM	2,000 kWh
Demand schedule 7–8 PM	2,000 kWh
Demand schedule 8–9 PM	2,000 kWh
Demand schedule 9–10 PM	2,000 kWh
Average demand schedule	9,000 kWh/day
Total annual load	3,285,000 kWh/yr

Table 8. PV system output power

Quantity	Value	Units
Rated capacity	2,200	kW
Mean output	373	kW
Mean output	8,940.79	kWh/d
Total production	3,263,389	kWh/yr
Hours of operation	4,369	hrs/yr
Levelized cost	0.039	\$/kWh

Figure 6. PV power output.



The capacities of the wind turbine generators are 4.5 MW and total electricity production is 3,424.742 MWh/yr and the mean output is 0.391 MW. The per unit cost of energy of the wind system is 0.04\$/kWh which is shown in Table 9 and hourly electricity production curve showed in Figure 7.

The rated capacity of the PEM electrolyzer is 3 MW. The Input power of the Electrolyzer is 6,499.506 MWh/yr which is supplied by solar & wind output power. Using sea water Electrolyzer produced H₂. Total Hydrogen is produced 1,400,659.79 kg/yr and the mean H₂ is 15.99 kg/yr. Tables 10 and 11 presents PEM electrolyzer input power and H₂ production output. Figure 8 shows electrolyzer input power. Figure 9 shows hydrogen production curve.

The rated capacity of the gas turbine generator is 3 MW. The operation schedule are (5–10) PM/day. Hydrogen consumed 137,803 L/yr. The production of the electricity is 3,281.028 MWh/yr which target peak load demand in patenga. Gas turbine generator output details Shows in Table 12 and Figure 10. Total Electricity Production and Hydrogen Consumption details shows in Tables 13 and 14. Figure 11 shows full system electricity production details.

Table 9. Wind turbine generator power

Quantity	Value	Units
Rated capacity	4,500	kW
Mean output	391	kW
Total production	3,424,742	kWh/yr
Hours of operation	8,319	hrs/yr
Levelized cost	0.040	\$/kWh

Figure 7. Wind turbine generator output.

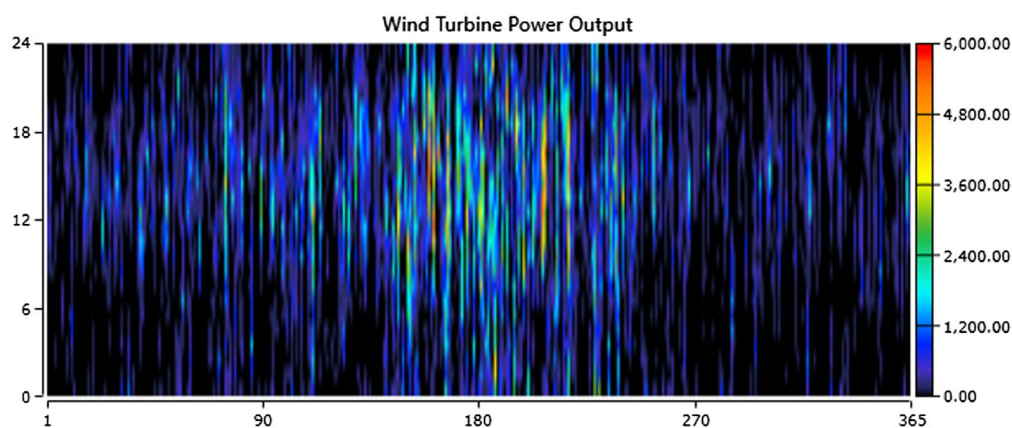


Table 10. PEM electrolyzer input power

Quantity	Value	Units
Rated capacity	3,000	kW
Mean output	741.95	kW
Total input power	6,499,506.36	kWh/yr
Hours of operation	8,458	hrs/yr

Table 11. H₂ production output

Quantity	Value	Units
H ₂ tank capacity	50,000	kg
Mean output	15.99	kg/hr
Total H ₂ production	1,400,659.79	kg/yr
Specific fuel consumption	46.41	kWh/kg

Figure 8. Electrolyzer input power.

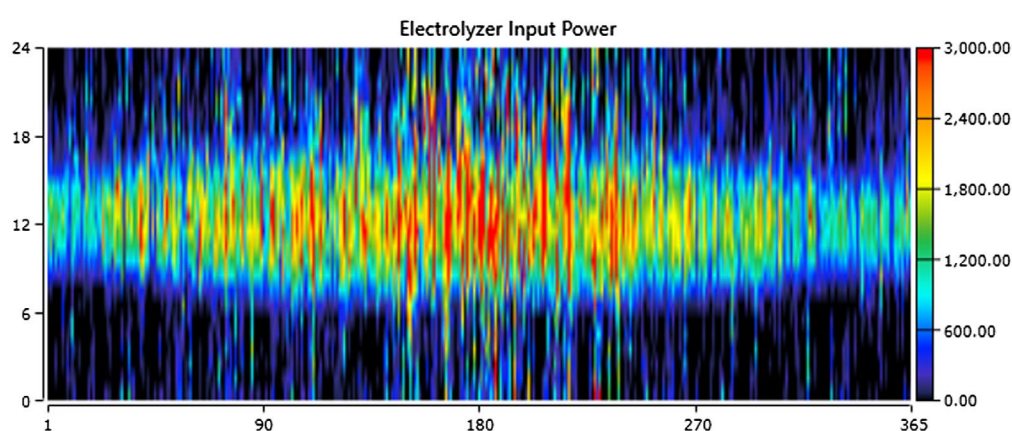


Figure 9. Hydrogen production.

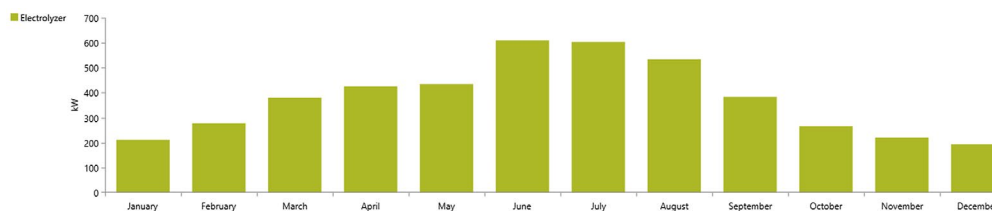


Table 12. Gas turbine 3 MW generator output

Quantity	Value	Units
Hours of operation	1,825	hrs/yr
Number of starts	365	starts/yr
Operational life	14	yr
Fixed generation cost	24.09	\$/hr
Electrical production	3,281,028	kWh/yr
Mean electrical output	1,798	kW
Fuel consumption	137,803	L/yr
Specific fuel consumption	0.04	L/kWh
Fuel energy input	4,593,439	kWh/yr
Mean electrical efficiency	71	%

Figure 10. Gas turbine generator output.



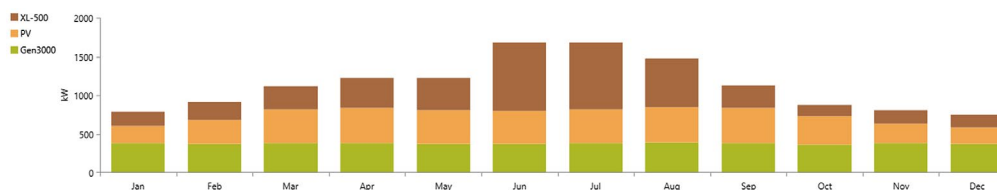
Table 13. Total electricity production

Component	Production (yr)	Fraction (%)
PV	3,263,389 kWh (dc)	33
Wind turbine generator	3,424,742 kWh (dc)	34
Gas turbine generator	3,281,028 kWh (ac)	33
Total	9,969,159 kWh	100

Table 14. Total hydrogen consumption

Total H ₂ production (H ₂)	140,060 kg/yr
Gas turbine generator consumption (H ₂)	137,803 kg/yr
Excess hydrogen	2,257 kg/yr
Load consumption	3,281,028 kWh

Figure 11. Total electricity production.



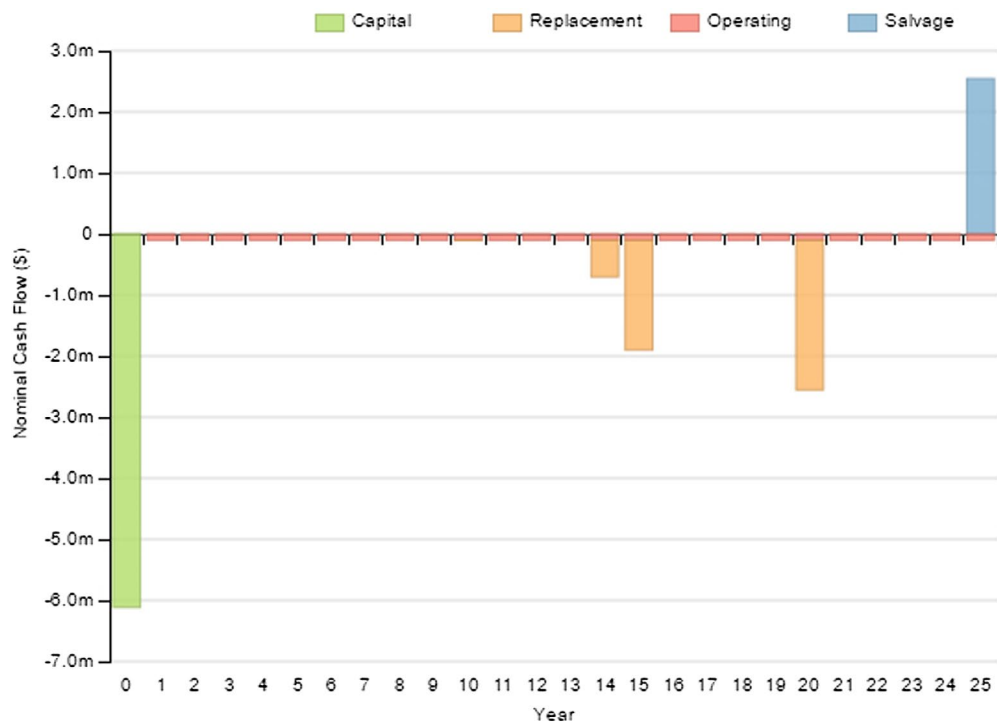
6. System costs

The initial cost of the system is very high about \$6,115,000. After 15 year Electrolyzer needs to replace and after 20 year also needs to replace PV cell & Wind turbine and total costs of the system are \$9,240,844. The annual cost is \$388,195. Table 15 and Figure 12 presents the overall system cost details.

Table 15. Overall system costs

Component	Initial capital (\$)	Replacement (\$)	Annual capital (\$)	Replacement (\$)
PV	1,760,000	509,658	111,729	32,354
Wind turbine generator	1,800,000	625,489	114,268	39,708
Electrolyzer	1,800,000	1,010,851	114,268	64,171
Hydrogen storage tank	5,000	915	317	58
Gas turbine generator	750,000	354,247	47,612	22,488
Total system	6,115,000	2,501,160	388,195	158,780
Overall cost of energy	\$0.178			

Figure 12. Overall system cost curve.



7. Conclusions

The proposed hybrid system will create new revolution in power sector. The system is completely green and won't emit any CO₂ & pollution and won't create any green house gas, also provides pure alternating current & won't provide any harmonics and distortion also doesn't need any power electronic devices like inverter which produce ripple factor. This hybrid system is designed for contributing in peak load demand in Kolatoli area in Cox's Bazar. The target load demand is 9 MWh/day and total demand in a year is 3,281.03 MWh/yr. To fulfill the target demand gas turbine generator needed 137,803 kg/yr and in this system produced hydrogen is about 140,060 kg/yr. The initial cost of the system is huge about \$6,115,000 and per unit **cost of energy of the system is 0.178\$/kWh** which is better than quick rental fuel based power plant in Bangladesh. So Access fuel could be used to transport & cooking sector.

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